Mathematically Grounded Evaluation of Large Language Models: A Survey and New Directions

Research Report
April 2026

Abstract

The evaluation of large language models (LLMs) has become a critical bottleneck in AI development. Despite rapid progress in model capabilities, evaluation methodology remains fragmented: static benchmarks saturate quickly, human evaluation is expensive and subjective, automated metrics correlate poorly with true quality, and production monitoring lacks formal guarantees. This paper bridges the empirical AI evaluation community, which has developed practical but mathematically informal methods, and the mathematical sciences, which offer powerful formal tools not yet systematically applied to LLM evaluation.

We provide three contributions. **Part I** presents a systematic survey of existing evaluation methods across 11 dimensions—knowledge, code, mathematics, generation quality, instruction following, safety, calibration, robustness, contamination detection, human preference, and compound systems—with consistent structure including mathematical foundations, comparison tables, and gap analysis for each dimension. **Part II** synthesizes six cross-cutting limitations: lack of stability guarantees, non-composable evaluations, absence of geometric drift detection, invisible structural failure modes, surface-level contamination detection, and no unifying framework. **Part III** proposes five novel mathematically grounded evaluation frameworks addressing these gaps: (1) topological drift detection via persistent homology, (2) sheaf-theoretic compositional evaluation with cohomological obstructions, (3) information-geometric evaluation manifolds with Fisher metric, (4) failure mode landscape analysis via TDA, and (5) spectral contamination detection via random matrix theory. Each framework includes formal definitions, key theorems, proof-of-concept implementations, and concrete paths to adoption. Our proof-of-concept experiments on MMLU performance data demonstrate the feasibility and informativeness of these mathematical approaches.

Keywords: Large language models, evaluation, persistent homology, sheaf theory, information geometry, random matrix theory, topological data analysis

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1 Introduction

Large language models (LLMs)—from GPT-4 [96] to Claude [7] to Gemini [50, 51] to LLaMA [41, 130, 131] and Mixtral [64]—have achieved remarkable capabilities across diverse tasks, from code generation [31] to mathematical reasoning [34] to creative writing. Chain-of-thought prompting [140] and self-consistency [136] have pushed reasoning capabilities further. Yet the methods we use to evaluate these systems have not kept pace with their development. The field faces a fundamental evaluation crisis characterized by several interrelated problems.

Holistic evaluation frameworks like HELM [75] attempt to address fragmentation, and embedding evaluation [91] has emerged as an important dimension. Dynamic benchmarks like Dynabench [67] and Chatbot Arena [32] resist saturation but introduce new challenges. Few-shot evaluation [21, 104] adds complexity by making results prompt-dependent.

Benchmark saturation. Static benchmarks saturate rapidly: MMLU [58] scores rose from 43.9% (GPT-3, 2020) to over 90% (GPT-o1, 2024) in just four years, approaching expert human performance of 89.8%. GSM8K [34] shows similar trends, with scores exceeding 97%. Once benchmarks saturate, they lose discriminative power, yet the community continues to rely on them.

Fragmented methodology. The evaluation landscape is fragmented across disconnected dimensions—knowledge benchmarks [58, 126], code evaluation [31], safety testing [151], calibration analysis [66], and human preference [32]—each with its own metrics, assumptions, and limitations. HELM [75] evaluates 42 scenarios across 7 metrics, but provides no formal framework for composing these into a consistent global assessment.

Lack of mathematical foundations. Most evaluation metrics are defined operationally without formal mathematical properties. We lack: (a) stability guarantees bounding how metrics change under perturbation, (b) compositionality enabling local evaluations to yield global assessments, (c) geometric frameworks for detecting distributional drift, and (d) spectral methods for identifying contamination at a structural level.

Contamination and gaming. Benchmark contamination threatens the validity of evaluation [124, 142, 143]. Current detection methods operate at the surface level (n-gram matching, embedding similarity) and can be evaded by rephrasing [143]. Holistic evaluation efforts [75, 109] highlight these limitations.

1.1 Contributions and Organization

This paper makes three main contributions:

1. **Comprehensive Survey (Part I, §2–§12).** We survey LLM evaluation across 11 dimensions with consistent structure: each section covers key methods, mathematical foundations, a comparison table, strengths, and limitations. We emphasize work from 2023–2025 while covering foundational earlier work.

2. **Gap Analysis (Part II, §13).** We identify six cross-cutting gaps in current evaluation methodology and connect each to formal mathematical concepts that could address it.

3. **Novel Frameworks (Part III, §14–§18).** We propose five mathematically grounded evaluation frameworks, each with formal definitions, key theorems, implementation considerations, and proof-of-concept results:

   - Topological drift detection via persistent homology (§14)
   - Sheaf-theoretic compositional evaluation (§15)

- Information-geometric evaluation manifolds (§16)
- Failure mode landscape analysis via TDA (§17)
- Spectral contamination detection via RMT (§18)

1.2 Related Work

Several surveys cover aspects of LLM evaluation. Chang et al. [27] provide a broad overview organized by capability type but lack mathematical analysis of evaluation foundations. Guo et al. [55] focus on downstream tasks and benchmarks. Biderman et al. [15] discuss practical reproducibility challenges. Huang et al. [62] survey hallucination detection. Xu et al. [142] survey contamination detection specifically.

On the mathematical side, Chazal and Michel [29] and Carlsson [25] introduce TDA for data science. Hensel et al. [60] survey TDA in machine learning. Nielsen [95] and Amari [4] provide foundations for information geometry. Martin and Mahoney [84] pioneer RMT analysis of neural networks. Hansen and Ghrist [56] and Bodnar et al. [16] apply sheaf theory to neural networks.

What distinguishes our work: No prior survey systematically connects mathematical tools (topology, sheaf theory, information geometry, spectral theory) to LLM evaluation gaps. We bridge the empirical evaluation community and mathematical sciences, providing both a comprehensive survey and actionable novel frameworks.

Table 1 provides unified notation used throughout.

Table 1: Unified notation used throughout this paper.

Symbol	**Meaning**
$\mathcal{M}$	Statistical manifold of model output distributions
$\mathcal{F}$	Sheaf over evaluation category
$\mathcal{X}$	Input space
$\mathcal{Y}$	Output/label space
$\mathcal{T}$	Set of evaluation tasks
$\mathcal{D}$	Dataset or distribution
θ	Model parameters
$p_\theta(y\|x)$	Model output distribution
$\mathrm{Dgm}_k(X)$	k-th persistence diagram of point cloud X
$d_B(\cdot,\cdot)$	Bottleneck distance between persistence diagrams
$W_p(\cdot,\cdot)$	p-Wasserstein distance between persistence diagrams
$d_{\mathrm{FR}}(\cdot,\cdot)$	Fisher-Rao distance
$\mathbf{F}(\theta)$	Fisher information matrix at θ
$H^k(\mathcal{U};\mathcal{F})$	k-th Čech cohomology of sheaf $\mathcal{F}$ on cover $\mathcal{U}$
λ_+, λ_-	Marchenko-Pastur upper/lower bounds
γ	Aspect ratio p/n for random matrix theory

Part I
Systematic Survey of LLM Evaluation Methods

2 Knowledge and Reasoning Evaluation

Knowledge and reasoning benchmarks assess an LLM's ability to recall factual information and apply logical reasoning across domains. This dimension has received the most attention, yet is also where benchmark saturation is most severe [18, 39].

2.1 Key Methods

MMLU [58] is the most widely used knowledge benchmark, comprising 15,908 multiple-choice questions across 57 subjects spanning STEM, humanities, social sciences, and professional domains. Questions are sourced from practice exams (GRE, USMLE, AP) at difficulty levels from elementary to professional. MMLU-Pro [137] extends MMLU with harder questions and 10 answer choices (vs. 4), reducing the ceiling effect and random-guess advantage.

BIG-Bench [126] provides 204 collaboratively designed tasks probing capabilities including logical reasoning, commonsense, and world knowledge. Notably, some tasks exhibit *emergent* behavior—near-random performance below a scale threshold followed by sharp improvement [139]. BIG-Bench Hard (BBH) [128] selects the 23 most challenging tasks where chain-of-thought [140] prompting shows the largest gains, providing a more discriminative subset.

ARC (AI2 Reasoning Challenge) [33] targets grade-school science questions requiring multi-step reasoning, while **HellaSwag** [144] evaluates commonsense reasoning through sentence completion with adversarially filtered distractors.

GPQA [114] (2024) provides graduate-level science questions verified by domain experts, achieving <35% accuracy for non-expert humans and providing a more challenging evaluation target. The emergence question has been debated: Schaeffer et al. [120] argue that apparent emergence may be an artifact of nonlinear metric choices.

Efficient evaluation. Recent work on tinyBenchmarks [108] and efficient benchmarking [106] shows that carefully selected subsets of 100 examples can approximate full benchmark results, addressing cost concerns. DyVal [149] proposes dynamic evaluation using graph-based test generation to resist contamination.

2.2 Mathematical Foundations

The primary metric is **accuracy**—the fraction of correctly answered questions. For multiple-choice with C choices, random baseline is $1/C$. Item Response Theory (IRT) provides a more sophisticated framework where item difficulty b_i and model ability θ are related by:

$$P(\text{correct} \mid \theta, b_i, a_i) = \frac{1}{1 + e^{-a_i(\theta - b_i)}} \tag{1}$$

where a_i is the item discrimination parameter. IRT enables adaptive testing and provides confidence intervals for ability estimates, but has seen limited adoption in LLM evaluation.

Table 2: Knowledge and reasoning benchmarks.

Benchmark	Size	Metric	Difficulty	Contamination Risk
MMLU	15.9K	Accuracy	Medium–High	High (public)
BIG-Bench	204 tasks	Task-specific	Variable	Medium
ARC	7.8K	Accuracy	Elementary	Medium
HellaSwag	10K	Accuracy	Medium	High
GPQA	448	Accuracy	Expert	Low (recent)

2.3 Comparison

2.4 Strengths and Limitations

Strengths include standardized comparison across models, broad domain coverage, and ease of automated evaluation. Key limitations include: (1) rapid saturation as models improve [2], (2) vulnerability to benchmark contamination [63, 143], (3) multiple-choice format artificially constraining evaluation to recognition rather than generation, (4) static nature preventing evaluation of evolving capabilities, and (5) sensitivity to prompt format [89]. The benchmark lottery phenomenon [39]—where model rankings change depending on which benchmark is used—further underscores the need for principled evaluation composition. Burnell et al. [24] argue for fundamentally rethinking how evaluation results are reported, including confidence intervals and effect sizes. IRT-based approaches [49] provide a more principled statistical framework but have seen limited adoption.

3 Code Generation Evaluation

3.1 Key Methods

HumanEval [31] contains 164 hand-crafted Python programming problems with function signatures, docstrings, and test cases. The **pass@k** metric provides an unbiased estimator:

$$\text{pass@k} = \mathop{\mathbb{E}}_{\text{Problems}} \left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}} \right] \tag{2}$$

where n is the number of samples and c the number that pass all tests.

MBPP [8] provides simpler function-level problems, while **CodeContests** [74] targets competitive programming. **DS-1000** [69] evaluates data science code generation with realistic library-specific tasks.

SWE-bench [65] evaluates real-world software engineering by requiring models to resolve actual GitHub issues, representing a significant increase in evaluation realism and the shift from function-level to repository-level evaluation.

3.2 Comparison

4 Mathematical Reasoning Evaluation

GSM8K [34] contains 8.5K grade-school math problems requiring 2–8 reasoning steps. Cobbe et al. introduced training *verifiers* (outcome-based reward models) that check intermediate reasoning steps. **Minerva** [71] demonstrated that training on mathematical data produces strong performance on MATH and GSM8K.

Table 3: Code generation benchmarks.

Benchmark	Metric	Scope	Difficulty
HumanEval	pass@k	Function synthesis	Medium
MBPP	pass@k	Function synthesis	Easy–Medium
SWE-bench	Resolve rate	Full repo	Hard
CodeContests	pass@k	Competitive	Very Hard

MATH [59] provides 12.5K competition-level problems across 7 subjects with difficulty levels 1–5. The **process reward model** approach [76] provides step-level supervision, achieving stronger performance than outcome-only verification. **MathVista** [80] evaluates mathematical reasoning in visual contexts.

The emergence of reasoning models (OpenAI o1 [97]) that use extended test-time compute has pushed GSM8K accuracy above 97%, effectively saturating this benchmark and motivating harder evaluation like AIME problems.

Evaluation uses exact match after normalization, but this is fragile to equivalent representations (e.g., $\frac{1}{2}$ vs. 0.5). Symbolic equivalence checking partially addresses this.

5 Language Generation Quality

Open-ended generation evaluation remains among the most challenging aspects of LLM evaluation. Traditional metrics like BLEU [101] and ROUGE [78] rely on surface-level n-gram matching and correlate poorly with human judgment of quality. This has motivated embedding-based and model-based approaches.

BERTScore [145] computes token-level cosine similarities between BERT embeddings of generated and reference texts:

$$\text{BERTScore} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_i \in \hat{x}} \max_{x_j \in x} \mathbf{x}_j^\top \hat{\mathbf{x}}_i \tag{3}$$

FActScore [88] decomposes generated text into atomic facts and verifies each against a knowledge source, providing fine-grained factuality evaluation.

MAUVE [107] measures the gap between neural text and human text distributions using KL divergence in a quantized embedding space. **BLEURT** [121] uses a learned metric fine-tuned on human judgments, achieving better correlation than BLEU.

Table 4: Generation quality metrics.

Metric	Basis	Correlation w/ Human	Reference-Free
BLEU	n-gram overlap	Low–Medium	No
ROUGE	n-gram recall	Low–Medium	No
BERTScore	Embedding similarity	Medium	No
FActScore	Atomic fact verification	High (factuality)	Yes
MAUVE	Distribution divergence	Medium–High	No

6 Instruction Following Evaluation

MT-Bench [146] uses 80 multi-turn questions graded by GPT-4, establishing the *LLM-as-a-judge* paradigm. Zheng et al. documented systematic biases: position bias (first response preferred), verbosity bias (longer responses preferred), and self-enhancement bias (models prefer their own outputs).

Chatbot Arena [32] provides the most ecologically valid evaluation through anonymous pairwise comparisons with real users, accumulating 240K+ votes. Rankings use the Bradley-Terry model:

$$P(\text{model } i \succ \text{model } j) = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}} \tag{4}$$

with statistical validity ensured via E-values from anytime-valid inference [134].

IFEval [147] provides verifiable instruction-following constraints (e.g., "write exactly 200 words"), enabling automated evaluation without LLM judges. **WildBench** [77] sources challenging tasks from real user interactions, while **Arena-Hard** [73] selects the most discriminative Arena questions for efficient evaluation. AlpacaFarm [42] provides a simulation framework for studying RLHF methods at lower cost. MT-Bench-101 [10] extends MT-Bench with fine-grained multi-turn dialogue capabilities.

Table 5: Instruction following evaluation methods.

Method	**Judge**	**Scale**	**Known Biases**
MT-Bench	LLM (GPT-4)	80 questions	Position, verbosity, self-enhancement
Chatbot Arena	Human crowd	240K+ votes	Demographic, task distribution
AlpacaEval	LLM	805 instructions	Length bias
IFEval	Automated	541 prompts	Limited to verifiable constraints

7 Safety and Alignment Evaluation

TruthfulQA [79] evaluates the tendency of models to produce false but plausible answers. **BBQ** [102] tests social bias through under-determined question answering. **DecodingTrust** [135] provides comprehensive trustworthiness evaluation across 8 dimensions: toxicity, stereotype bias, adversarial robustness, out-of-distribution robustness, privacy, machine ethics, fairness, and hallucination.

Constitutional AI [11] introduced self-improvement for harmlessness where an AI critiques and revises its own outputs based on a set of principles (the "constitution").

Adversarial safety evaluation has advanced rapidly with **GCG** [151], which generates universal adversarial suffixes via gradient-based optimization that transfer across models, demonstrating fundamental fragility of alignment. **AutoDAN** [150] generates human-readable adversarial prompts, while **PAIR** [28] uses an attacker LLM to iteratively refine jailbreaks in only 20 queries. HarmBench [85] provides a standardized framework for comparing attack and defense methods. Red teaming can be automated using language models themselves [105].

8 Calibration and Uncertainty Evaluation

Expected Calibration Error (ECE) partitions predictions into B bins by confidence and measures the gap between confidence and accuracy:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} |\text{acc}(S_b) - \text{conf}(S_b)| \tag{5}$$

Table 6: Safety evaluation approaches.

Method	Dimensions	Automated	Adversarial
TruthfulQA	Truthfulness	Yes	No
BBQ	Social bias	Yes	Partially
DecodingTrust	8 dimensions	Yes	Yes
GCG Attack	Jailbreaking	Yes	Yes
Red Teaming	Open-ended	Partially	Yes

Kadavath et al. [66] showed LLMs exhibit reasonable calibration on multiple-choice tasks, improving with scale. They introduced P(True) and P(IK) ("I Know") self-evaluation metrics. The foundational work of Guo et al. [54] on neural network calibration and Naeini et al. [93] on ECE provide the methodological basis. Shen et al. [123] further investigate LLM self-knowledge.

Conformal prediction [6, 127] provides distribution-free coverage guarantees: given a calibration set and error rate α, conformal prediction constructs prediction sets $C(x)$ satisfying $P(y \in C(x)) \geq 1 - \alpha$ regardless of the underlying distribution. Kumar et al. [68] apply conformal prediction specifically to LLM multiple-choice QA.

E-values [134] provide anytime-valid statistical inference, used in Chatbot Arena [32] for ranking validity.

9 Robustness and Adversarial Evaluation

The GCG attack [151] optimizes adversarial suffixes via:

$$\min_{\delta \in \mathcal{V}^L} -\log p_\theta(y_{\text{target}} \mid x \oplus \delta) \tag{6}$$

where δ is a sequence of tokens from vocabulary $\mathcal{V}$, optimized by greedy coordinate gradient. These suffixes *transfer* across models including black-box APIs, revealing shared vulnerabilities. This has deep connections to the topology of the safety boundary in token space.

Robustness evaluation also includes distribution shift analysis [110, 113]: models may perform well on in-distribution benchmarks but degrade on shifted inputs. Multi-prompt evaluation [89] tests sensitivity to prompt phrasing.

10 Contamination Detection

MIN-K% PROB [124] provides a reference-free membership inference test: for text $x = (x_1, \ldots, x_T)$, select the $k\%$ of tokens with lowest model probability and average their log-likelihoods:

$$\text{MIN-K\%}(x) = \frac{1}{|S_k|} \sum_{t \in S_k} \log p_\theta(x_t \mid x_{<t}) \tag{7}$$

where S_k contains the $k\%$ lowest-probability tokens. The hypothesis is that unseen text has more outlier low-probability tokens.

Yang et al. [143] demonstrated that n-gram decontamination is insufficient: LLM-rephrased test samples bypass detection but still inflate benchmark scores (a 13B model trained on rephrased MMLU achieves 85.9% vs. GPT-4's 86.4%). Xu et al. [142] provide a comprehensive survey of contamination detection methods, while Oren et al. [99] propose a method for proving contamination in black-box models. Deng et al. [40] and Sainz et al. [119] provide additional detection methodologies. Jacovi et al. [63] advocate for encrypting test data to prevent contamination entirely.

Table 7: Contamination detection methods.

Method	Access Required	Detects Rephrasing	Formal Guarantee
n-gram overlap	Training data	No	None
Embedding similarity	Embeddings	Partially	None
MIN-K% PROB	Log-probs	Partially	Hypothesis test
LLM Decontaminator	API access	Yes	None
Temporal analysis	Multiple versions	N/A	Statistical

11 Human Preference Evaluation

The Bradley-Terry model (4) underlies both Chatbot Arena rankings and RLHF training [100]. DPO [111] reformulates RLHF by directly optimizing the policy using the implicit reward:

$$r^*(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)} + \beta \log Z(x) \tag{8}$$

This eliminates the need for an explicit reward model while maintaining the Bradley-Terry preference structure.

The foundational Bradley-Terry model [19] dates to 1952, while the Elo system [44] was designed for chess. Both assume *transitive* preferences, which may not hold for LLMs. Ethayarajh et al. [47] propose KTO as an alternative using prospect-theoretic utility. Starling-7B [148] demonstrates that RLAIF (AI feedback) can approach RLHF quality.

Ethayarajh and Jurafsky [46] raise fundamental questions about whether LLM evaluators truly measure what they claim. The connection between Bradley-Terry models and exponential families makes this dimension particularly amenable to information-geometric analysis (§16).

12 RAG and Compound System Evaluation

RAGAS [45] evaluates RAG pipelines along four dimensions: faithfulness, answer relevance, context precision, and context recall. **ARES** [118] uses prediction-powered inference to provide confidence intervals for RAG evaluation metrics.

The original RAG framework [70] established the paradigm, and evaluation has since expanded to long-context settings [12].

The fundamental challenge is *compositionality*: retrieval quality and generation quality interact non-linearly, and component-wise metrics do not compose into reliable end-to-end assessments. This motivates the sheaf-theoretic approach of §15.

12.1 Hallucination Evaluation

Hallucination detection is critical for RAG and generation tasks. Huang et al. [62] provide a comprehensive taxonomy of hallucination types. Li et al. [72] introduce HaluEval, a large-scale benchmark. SelfCheckGPT [81] detects hallucinations without external knowledge by checking consistency across multiple generations. SimpleQA [98] provides factuality evaluation via short-form QA.

Part II
Cross-Cutting Gap Analysis

13 Six Fundamental Gaps in LLM Evaluation

Our survey reveals six cross-cutting limitations that span multiple evaluation dimensions. These are not minor technical issues but fundamental gaps in the mathematical foundations of evaluation methodology.

13.1 Gap 1: Lack of Formal Stability Guarantees

Current evaluation metrics—accuracy, Elo rating, ECE, pass@k—are defined operationally without formal bounds on how they change under perturbation. When the input distribution shifts slightly (e.g., paraphrased questions, different demographic groups), we have no guarantee on how much the metric can change. This contrasts with, e.g., Lipschitz-continuous functions in analysis, where perturbation bounds are fundamental.

Mathematical remedy: The stability theorem of persistent homology [29] guarantees that small perturbations in point cloud data produce bounded changes in persistence diagrams:

$$d_B(\mathrm{Dgm}(X), \mathrm{Dgm}(Y)) \leq d_{\mathrm{GH}}(X, Y) \tag{9}$$

where d_B is the bottleneck distance and d_{GH} is the Gromov-Hausdorff distance.

13.2 Gap 2: Non-Composable Evaluations

HELM [75] evaluates 42 scenarios across 7 metrics, producing a matrix of scores. But how should these be combined? Current practice uses ad-hoc aggregation (averaging, ranking) without formal guarantees that local evaluations compose into a consistent global assessment. When two evaluation contexts overlap (e.g., reasoning evaluated in both math and science contexts), there is no formal requirement that their assessments agree on the overlap.

Mathematical remedy: Sheaf theory [20] provides exactly this: a sheaf assigns data to open sets (evaluation contexts) with *restriction maps* ensuring consistency, and the *gluing axiom* guarantees that consistent local sections compose into unique global sections. Sheaf cohomology measures the obstruction to gluing.

13.3 Gap 3: No Geometric Framework for Distributional Drift

When a model is updated (fine-tuning, RLHF, new data), or when evaluation shifts to a new domain, there is no principled way to measure how the output distribution has changed. KL divergence is commonly used but is not a true metric (asymmetric, unbounded) and does not provide a natural geometry.

Mathematical remedy: The Fisher-Rao distance [95] provides the unique Riemannian metric on statistical manifolds that is invariant under sufficient statistics. It gives a proper metric structure for measuring distributional changes.

13.4 Gap 4: Invisible Structural Failure Modes

Aggregate metrics (mean accuracy, overall pass rate) obscure systematic patterns of correlated failures. A model might fail on all questions requiring multi-step temporal reasoning while succeeding on

similar-looking but structurally simpler questions. Pointwise metrics cannot detect such topological structure in failure patterns.

Mathematical remedy: Persistent homology applied to failure patterns reveals multi-scale structure—clusters of correlated failures, cycles of failure relationships—that aggregate statistics miss.

13.5 Gap 5: Surface-Level Contamination Detection

Current contamination detection operates at the surface level: n-gram overlap, embedding similarity, or token-level likelihood [124]. Yang et al. [143] showed these are easily evaded by paraphrasing. We need detection methods that identify contamination at a structural level.

Mathematical remedy: Random matrix theory [82, 84] predicts that the eigenvalue distribution of performance correlation matrices follows the Marchenko-Pastur distribution under the null hypothesis of no contamination. Systematic contamination creates outlier eigenvalues beyond the Tracy-Widom threshold [132]. The BBP phase transition [13] characterizes when signal eigenvalues become detectable above the noise floor, and Wigner's semicircle law [141] provides baseline predictions for symmetric matrices.

13.6 Gap 6: No Unifying Mathematical Framework

Each evaluation dimension uses its own ad-hoc metrics without a common mathematical language. This prevents: cross-dimensional comparison, formal composition, unified quality assurance, and principled metric design. A unifying framework would enable evaluation as a mathematical discipline with formal guarantees.

Part III
Novel Mathematically Grounded Evaluation Frameworks

14 Topological Drift Detection via Persistent Homology

14.1 Motivation

Model capabilities evolve through training, fine-tuning, and deployment. Current evaluation captures snapshots (accuracy at time t) but cannot characterize the *structural* changes in capability profiles. We propose using persistent homology [43, 52] to track the topological evolution of model capability spaces, leveraging the stability theorem [30, 35] to provide formal guarantees on drift detection. This builds on foundational work in TDA [25, 29] and its applications to neural networks [26, 94, 115].

14.2 Formal Setup

Definition 14.1 (Capability Point Cloud)**.** *Given a set of n models $\{M_1, \ldots, M_n\}$ and m evaluation tasks $\{T_1, \ldots, T_m\}$, the* capability point cloud *is $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^m$ where $x_i = (score(M_i, T_1), \ldots, score(M_i, T_m))$.*

Definition 14.2 (Vietoris-Rips Complex)**.** *For point cloud X and scale parameter $\epsilon > 0$, the* Vietoris-Rips complex $\mathrm{VR}_\epsilon(X)$ *is the simplicial complex where a k-simplex $\{x_{i_0}, \ldots, x_{i_k}\}$ is included if and only if $d(x_{i_j}, x_{i_l}) \leq \epsilon$ for all $0 \leq j, l \leq k$.*

As ϵ increases from 0 to ∞, topological features (connected components, loops, cavities) appear (are "born") and disappear ("die"). The *persistence diagram* $\mathrm{Dgm}_k(X)$ records these birth-death pairs for k-dimensional homology.

Theorem 14.3 (Stability [29])**.** *For point clouds X and Y,*

$$d_B(\mathrm{Dgm}_k(X), \mathrm{Dgm}_k(Y)) \leq d_{\mathrm{GH}}(X, Y) \leq 2d_H(X, Y) \tag{10}$$

where d_H is the Hausdorff distance. This holds for all dimensions $k \geq 0$.

Corollary 14.4 (Drift Detection Guarantee)**.** *If models at times t_1 and t_2 produce capability point clouds X_1 and X_2 with $d_H(X_1, X_2) \leq \delta$, then the topological features change by at most 2δ in bottleneck distance. Conversely, if $d_B(\mathrm{Dgm}_k(X_1), \mathrm{Dgm}_k(X_2)) > \tau$, then $d_H(X_1, X_2) > \tau/2$—the capability landscape has genuinely changed.*

14.3 Drift Detection Algorithm

Algorithm 1 Topological Drift Detection

Require: Capability point clouds X_1, X_2 at times t_1, t_2; threshold τ
Ensure: Drift detected (boolean), drift magnitude

```
Compute persistence diagrams D_1 ← Dgm(X_1), D_2 ← Dgm(X_2)
for dimension k = 0, 1, ... do
    δ_k ← W_p(D_1^(k), D_2^(k)) {Wasserstein distance}
    β_k ← d_B(D_1^(k), D_2^(k)) {Bottleneck distance}
end for
Drift magnitude ← max_k β_k
return max_k β_k > τ, drift magnitude
```

14.4 Proof-of-Concept Results

We applied persistent homology to simulated capability profiles of 12 LLMs across 57 MMLU subjects. The point cloud in $\mathbb{R}^{57}$ was analyzed using the Vietoris-Rips complex (computed via Ripser [14]).

H0 analysis: The persistence diagram revealed 12 connected components that merge into one as the filtration parameter increases, with the longest-lived features corresponding to the gap between weak models (Mistral 7B, LLaMA-3 8B) and strong models (GPT-4o, Claude-3.5 Sonnet). The most persistent H_0 features capture genuine capability clusters.

Temporal drift: Comparing early (2023) and late (2024) model cohorts, we measured a Wasserstein distance of 1.77 and bottleneck distance of 0.68 for H_0 features. By the stability theorem, this lower-bounds the Hausdorff distance between the capability landscapes of the two generations at ≥ 0.34, indicating genuine structural change.

Vectorization. For integration with machine learning pipelines, persistence diagrams can be vectorized using persistence images [1], persistence landscapes [22], or perturbation-robust representations [125]. Statistical analysis of persistence diagrams uses probability measures on diagram space [87] and Fréchet means [133].

Connections to neural network analysis. Rieck et al. [115] introduced neural persistence as a complexity measure via persistent homology of network weight graphs. Corneanu et al. [36] showed persistence diagrams can predict generalization without test data. Watanabe and Yamana [138] applied topological measures to deep network evaluation. Topology layers [48] and topological autoencoders [90] enable end-to-end topological optimization. Hofer et al. [61] use persistent homology for connectivity-optimized representations. Ramamurthy et al. [112] apply TDA to decision boundary analysis.

Computational considerations: Ripser [14] computes Vietoris-Rips persistence in $O(2^n)$ worst case but is highly optimized for practical inputs. GUDHI [53] and Giotto-TDA [129] provide comprehensive Python interfaces. For our 12-model, 57-subject point cloud, computation took <1 second on CPU. Seversky et al. [122] demonstrate TDA for time-series analysis, relevant to temporal drift detection.

15 Sheaf-Theoretic Compositional Evaluation

15.1 Motivation

Modern LLM evaluation involves multiple overlapping evaluation contexts: the same model is evaluated on math within a safety benchmark and within a reasoning benchmark. Current practice treats these independently, but consistency across overlapping contexts is desirable and formally enforceable via sheaf theory [20, 37]. Robinson [116, 117] established sheaves as the canonical data structure for sensor integration, providing a direct precedent for evaluation composition.

15.2 Formal Setup

Definition 15.1 (Evaluation Category)**.** *Let $\mathcal{C}$ be a category whose objects are evaluation contexts (tasks, domains, metrics) and whose morphisms are refinement relations (e.g., "STEM" refines to "mathematics" and "physics").*

Definition 15.2 (Evaluation Sheaf)**.** *An* evaluation sheaf *$\mathcal{F}$ on $\mathcal{C}$ assigns to each evaluation context $U \in \mathrm{Ob}(\mathcal{C})$:*

1. *A set $\mathcal{F}(U)$ of evaluation measurements (the "sections" over U);*
2. *For each morphism $U \to V$ (refinement), a restriction map $\rho_{V,U} : \mathcal{F}(U) \to \mathcal{F}(V)$*

satisfying: (a) ***Identity:*** *$\rho_{U,U} = \mathrm{id}$; (b)* ***Composition:*** *$\rho_{W,V} \circ \rho_{V,U} = \rho_{W,U}$; and the sheaf conditions: (c)* ***Locality:*** *if two global sections agree on all refinements, they are equal; (d)* ***Gluing:*** *if local sections agree on overlaps, they glue to a unique global section.*

Definition 15.3 (Cohomological Obstruction)**.** *Given an open cover $\mathcal{U} = \{U_i\}$ of evaluation contexts and a presheaf $\mathcal{F}$, the* Čech cohomology *$H^k(\mathcal{U}; \mathcal{F})$ measures the obstruction to extending local evaluations to global ones. Specifically:*

- *$H^0(\mathcal{U}; \mathcal{F})$ is the space of global sections (consistent global evaluations);*
- *$H^1(\mathcal{U}; \mathcal{F}) \neq 0$ indicates that local evaluations* cannot *be consistently composed.*

Proposition 15.4 (Evaluation Consistency Criterion)**.** *An evaluation framework $\mathcal{F}$ is* globally consistent *if and only if $H^1(\mathcal{U}; \mathcal{F}) = 0$ for all open covers $\mathcal{U}$ of the evaluation category. When $H^1 \neq 0$, the dimension of H^1 quantifies the degree of inconsistency.*

15.3 Construction for HELM-Style Evaluation

We construct a concrete sheaf for multi-metric evaluation. Let the base space be a graph $G = (V, E)$ where:

- Vertices V include a global node, category nodes (STEM, Humanities, etc.), and individual task nodes;
- Edges E connect each category to its tasks and the global node to categories.

The sheaf assigns to each vertex v the vector space $\mathcal{F}(v) = \mathbb{R}^d$ of d-dimensional evaluation measurements, and to each edge $e = (u, v)$ a linear restriction map $\mathcal{F}_e : \mathcal{F}(u) \to \mathcal{F}(v)$ (e.g., averaging over subcategories).

Sheaf Laplacian. The *sheaf Laplacian* $L_{\mathcal{F}}$ generalizes the graph Laplacian and is defined via the coboundary operator $\delta : C^0(G; \mathcal{F}) \to C^1(G; \mathcal{F})$. Its kernel equals $H^0(G; \mathcal{F})$ (global sections), and its spectral properties reveal evaluation consistency: small eigenvalues indicate near-consistent evaluations. This extends neural sheaf diffusion [16] and sheaf neural networks [56] from GNN contexts to evaluation. The connection to persistent homology via the persistence map [38] enables joint topological-sheaf analysis. Algebraic topology [57, 92] provides the rigorous foundations.

15.4 Proof-of-Concept Results

We constructed a sheaf over a task decomposition graph with 63 nodes (1 global, 5 categories, 57 subjects) and 65 edges. For each of 12 models, we computed the consistency ratio (between-category variance / within-category variance):

Models with high consistency ratios (Gemini 1.5 Pro: 1.83, Mixtral: 1.72) exhibit heterogeneous capability profiles where category-level summaries lose substantial information. Models with low ratios (LLaMA-3 8B: 0.48) have more uniform profiles where aggregation is less lossy. The sheaf-theoretic perspective reveals that standard averaging hides meaningful structure for models with high consistency ratios.

16 Information-Geometric Evaluation Manifolds

16.1 Motivation

The Bradley-Terry model (4) underlying Chatbot Arena rankings forms an *exponential family*—a class of distributions with rich geometric structure [4, 9]. By equipping the parameter space with the Fisher information metric, we obtain a Riemannian manifold where geodesic distances measure genuine distributional differences that raw parameter differences miss. The natural gradient [3, 83] exploits this geometry for optimization; we repurpose it for evaluation.

16.2 Formal Setup

Definition 16.1 (Statistical Manifold)**.** *A* statistical manifold $(\mathcal{M}, g)$ *is a smooth manifold* $\mathcal{M}$ *of probability distributions equipped with the* Fisher information metric*:*

$$g_{ij}(\theta) = \mathbf{F}_{ij}(\theta) = \mathbb{E}_{p_\theta}\left[\frac{\partial \log p_\theta}{\partial \theta_i} \cdot \frac{\partial \log p_\theta}{\partial \theta_j}\right] \tag{11}$$

Theorem 16.2 (Čencov, 1982)**.** *The Fisher information metric is, up to a constant, the unique Riemannian metric on statistical manifolds that is invariant under sufficient statistics.*

Definition 16.3 (Fisher-Rao Distance)**.** *The* Fisher-Rao distance *between distributions p_{θ_1} and p_{θ_2} is the geodesic distance on $(\mathcal{M}, g)$:*

$$d_{\mathrm{FR}}(p_{\theta_1}, p_{\theta_2}) = \inf_{\gamma} \int_0^1 \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))}\, dt \tag{12}$$

where the infimum is over smooth paths $\gamma : [0, 1] \to \mathcal{M}$ with $\gamma(0) = \theta_1$, $\gamma(1) = \theta_2$.

The Fisher-Rao distance belongs to the family of f-divergences [23] and is the unique separable divergence satisfying information monotonicity. For categorical distributions on the probability simplex Δ^{n-1}, the Fisher-Rao distance equals twice the Bhattacharyya angle:

$$d_{\mathrm{FR}}(p, q) = 2 \arccos\left(\sum_{i=1}^{n} \sqrt{p_i q_i}\right) \tag{13}$$

Proposition 16.4 (Evaluation Drift Detection)**.** *Let p_t denote the output distribution of a model at time t on a fixed evaluation set. The evaluation drift between times t_1 and t_2 is:*

$$\Delta_{drift}(t_1, t_2) = d_{\mathrm{FR}}(p_{t_1}, p_{t_2}) \tag{14}$$

This is a proper metric (symmetric, satisfies triangle inequality) and is invariant under reparameterization—unlike KL divergence.

16.3 Application to Bradley-Terry Rankings

The Bradley-Terry model (4) parameterizes win probabilities as a multinomial logistic model. The Fisher information matrix for n models has entries:

$$\mathbf{F}_{ij} = \begin{cases} \sum_{k \neq i} n_{ik} \pi_{ik}(1 - \pi_{ik}) & \text{if } i = j \\ -n_{ij} \pi_{ij}(1 - \pi_{ij}) & \text{if } i \neq j \end{cases} \tag{15}$$

where $\pi_{ij} = P(i \succ j)$ and n_{ij} is the number of comparisons between i and j.

The Fisher information trace $\mathrm{tr}(\mathbf{F})$ measures the total sensitivity of the ranking to parameter perturbation. Rankings with low Fisher information are *robust*; those with high Fisher information are *unstable*—a small change in votes could substantially alter rankings.

16.4 Proof-of-Concept Results

We constructed capability distributions for 12 models by discretizing their 57-subject MMLU performance into 10-bin histograms, then computed the full Fisher-Rao distance matrix using (13).

The maximum Fisher-Rao distance was 2.82 (between Mistral 7B and GPT-4o), and the mean pairwise distance was 1.46. MDS embedding of the Fisher-Rao distance matrix reveals a clear separation between strong and weak models, with interesting intermediate structure: Claude-3 Opus and GPT-4 are geometrically close despite different raw accuracies, suggesting similar distributional profiles.

The Fisher information trace analysis reveals that weaker models (Mistral 7B, GPT-3.5 Turbo) have higher Fisher information—their performance distributions are more concentrated (peaked), making them more sensitive to perturbations. Stronger models have flatter, more spread distributions and are geometrically more stable.

17 Failure Mode Landscape Analysis via TDA

17.1 Motivation

Aggregate evaluation metrics hide structural patterns in model failures. A model with 80% accuracy might fail uniformly at random, or it might systematically fail on a correlated cluster of tasks. These scenarios have different implications for deployment but are invisible to accuracy alone.

17.2 Formal Setup

Definition 17.1 (Failure Profile)**.** *For model M and threshold τ, the* failure profile *is $f_M^\tau = (f_1, \ldots, f_m) \in \{0,1\}^m$ where $f_j = \mathbf{1}[score(M, T_j) < \tau]$.*

We equip the space of failure profiles with the Hamming metric and compute persistent homology of the resulting point cloud.

Proposition 17.2 (Multi-Scale Failure Structure)**.** *The persistence diagram $\mathrm{Dgm}_k(\{f_M^\tau\}_M)$ at threshold τ reveals:*

- *H_0: Clusters of models with similar failure patterns (connected components);*
- *H_1: Cyclic failure relationships where model A fails where B succeeds, B fails where C succeeds, and C fails where A succeeds (1-dimensional holes).*

Definition 17.3 (Failure Persistence Landscape)**.** *Varying τ from 0 to 1, the* failure persistence landscape *is the function $\tau \mapsto \mathrm{Dgm}(\{f_M^\tau\}_M)$. Phase transitions in this landscape (sudden appearance/disappearance of persistent features) indicate critical capability thresholds.*

17.3 Proof-of-Concept Results

Analyzing 12 models across 57 MMLU subjects at thresholds $\tau \in \{0.5, 0.6, 0.7, 0.8\}$:

At $\tau = 0.5$, failure rates are low (mean 3.9%) and all models form a single connected component ($\beta_0 = 1$). At $\tau = 0.7$, failure rates increase (mean 21.1%) but models still cluster together. At $\tau = 0.8$, a phase transition occurs: failure rates reach 54.8%, and H_1 features emerge ($\beta_1 = 2$), indicating cyclic failure relationships between model groups.

The subject failure correlation matrix at $\tau = 0.7$ reveals blocks of correlated subjects, confirming that failures are structured rather than random.

18 Spectral Contamination Detection via Random Matrix Theory

18.1 Motivation

Benchmark contamination inflates model performance in ways that surface-level detection misses after rephrasing [143]. We propose detecting contamination through the *spectral signature* it leaves in performance correlation matrices, using predictions from random matrix theory [5, 86] as the null hypothesis. This extends the seminal work of Martin and Mahoney [84] on spectral analysis of neural network weight matrices, and Paul and Roberts [103] on transformer-specific spectral signatures. RMT has proven powerful in finance [17] for detecting signal from noise in correlation matrices—we apply the same principle to evaluation.

18.2 Formal Setup

Definition 18.1 (Performance Matrix)**.** *For n models and p tasks, the* performance matrix $X \in \mathbb{R}^{n \times p}$ *has entries $X_{ij} = score(M_i, T_j)$. The* sample covariance matrix *is $C = \frac{1}{n}\tilde{X}^T\tilde{X}$ where $\tilde{X}$ is centered and standardized.*

Theorem 18.2 (Marchenko-Pastur Law [82])**.** *Let $X \in \mathbb{R}^{n \times p}$ have i.i.d. entries with mean 0 and variance σ^2, with $p/n \to \gamma > 0$ as $n, p \to \infty$. The empirical spectral distribution of $C = \frac{1}{n}X^TX$ converges to the Marchenko-Pastur distribution with density:*

$$f_{\mathrm{MP}}(\lambda) = \frac{\gamma}{2\pi\sigma^2\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \cdot \mathbf{1}_{[\lambda_-, \lambda_+]}(\lambda) \tag{16}$$

where $\lambda_\pm = \sigma^2(1 \pm \sqrt{1/\gamma})^2$.

Theorem 18.3 (Tracy-Widom Fluctuation)**.** *Under the Marchenko-Pastur null, the largest eigenvalue $\lambda_{\max}$ satisfies:*

$$\frac{\lambda_{\max} - \lambda_+}{n^{-2/3}\lambda_+} \xrightarrow{d} TW_1 \tag{17}$$

where TW_1 is the Tracy-Widom distribution of type 1.

Definition 18.4 (Contamination Score)**.** *The* spectral contamination score *is:*

$$S_{contam} = \frac{|\{\lambda_i : \lambda_i > \lambda_+ + c \cdot n^{-2/3}\lambda_+\}|}{p} \tag{18}$$

where c is chosen to achieve a desired false positive rate under TW_1.

Proposition 18.5 (Contamination Detection via Spectral Outliers)**.** *Systematic contamination affecting k tasks creates a rank-k signal matrix added to the noise, producing k eigenvalues beyond the Marchenko-Pastur support. When the signal strength exceeds $\sigma^2/\sqrt{\gamma}$ (the BBP phase transition), these outlier eigenvalues are detectable.*

18.3 Leverage Score Identification

Once outlier eigenvalues are detected, the corresponding eigenvectors identify contaminated tasks through their *leverage scores*:

$$\ell_j = \sum_{i:\lambda_i > \lambda_+} v_{ij}^2 \tag{19}$$

where v_i is the eigenvector associated with outlier eigenvalue λ_i. Tasks with high leverage scores are disproportionately responsible for the spectral anomaly.

18.4 Proof-of-Concept Results

We analyzed the performance matrix of 12 models on 57 MMLU subjects ($\gamma = 57/12 = 4.75$). The Marchenko-Pastur bounds are $\lambda_- = 0.29$ and $\lambda_+ = 2.13$.

Clean data: 4 eigenvalues exceed λ_+, with a Tracy-Widom statistic of 83.1, indicating genuine structure beyond noise (model capability differences, subject difficulty correlations). This is expected—real performance data is not pure noise.

Contaminated data: After injecting systematic performance boosts on 10 randomly selected subjects (simulating contamination), the eigenvalue distribution shifts: the top eigenvalues increase,

and the spectral profile becomes more outlier-heavy. The contaminated subjects appear among the top leverage score subjects, demonstrating that spectral analysis can localize contamination.

Eigenvalue spacing: The normalized eigenvalue spacing distribution falls between the Wigner surmise (GOE, indicating correlated structure) and Poisson distribution (independent), suggesting moderate inter-task correlations in the clean data.

19 Conclusion and Future Work

We have presented a comprehensive survey of LLM evaluation across 11 dimensions, identified six fundamental gaps in current methodology, and proposed five novel mathematically grounded frameworks addressing these gaps. Our key findings:

1. **The evaluation landscape is fragmented** across dimensions with no unifying mathematical language, preventing formal composition and cross-dimensional analysis.
2. **Stability guarantees are absent** from every major evaluation framework, leaving open the question of how much metrics can change under natural perturbations.
3. **Mathematical tools exist** in topology, algebraic geometry, information geometry, and spectral theory that directly address these gaps, with established theoretical foundations and practical implementations.
4. **Proof-of-concept implementations** demonstrate feasibility: persistent homology reveals capability structure in MMLU profiles, sheaf-theoretic analysis quantifies evaluation consistency, Fisher-Rao distances provide proper geometry for model comparison, TDA reveals failure mode clustering, and spectral analysis detects contamination signatures.

19.1 Future Directions

Scaling to production. Our proof-of-concept operates on simulated and small-scale data. Scaling to real evaluation scenarios (thousands of models, millions of queries) requires computational advances in TDA (approximate persistence) and efficient sheaf computation.

Theoretical development. The sheaf-theoretic framework needs full development of the cohomological obstruction theory, including computability results and connections to model-theoretic evaluation. Information-geometric analysis of Bradley-Terry manifolds could yield optimal evaluation designs.

Empirical validation. Each proposed framework should be validated against known ground truth: artificially contaminated benchmarks (for spectral methods), intentionally modified models (for drift detection), and evaluation datasets with known inconsistencies (for sheaf analysis).

Integration. The five frameworks are complementary: topological methods provide stability, sheaf theory provides compositionality, information geometry provides drift detection, TDA provides failure analysis, and RMT provides contamination detection. A unified evaluation platform integrating all five would represent a major advance.

Category-theoretic unification. The sheaf-theoretic approach hints at a deeper category-theoretic framework where evaluation dimensions, metrics, and models form a category, and evaluation is a functor. This could provide the unifying mathematical language the field currently lacks.

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